

260 LAB Report 08

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Group 05

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Name of the experiment: Design and Implementation of the following circuit.

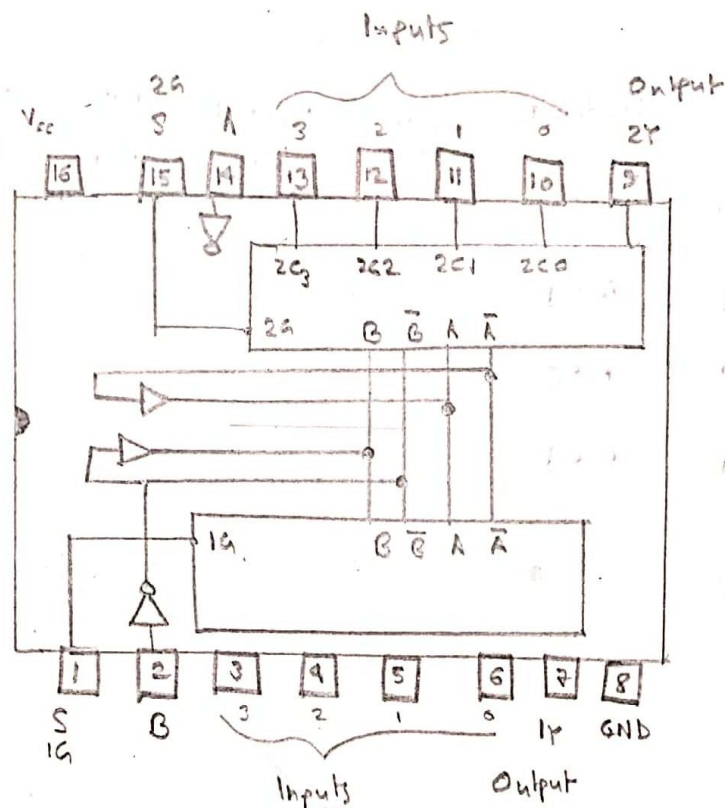
Objective:

- I. To understand how MUX and P.Adder works.
- II. Can build a circuit using the principles of MUX & parallel Adder which takes four 2 bit numbers as input and adds up two of them following select pins combination using 2 bit P. Adder.

Required components and Equipments:

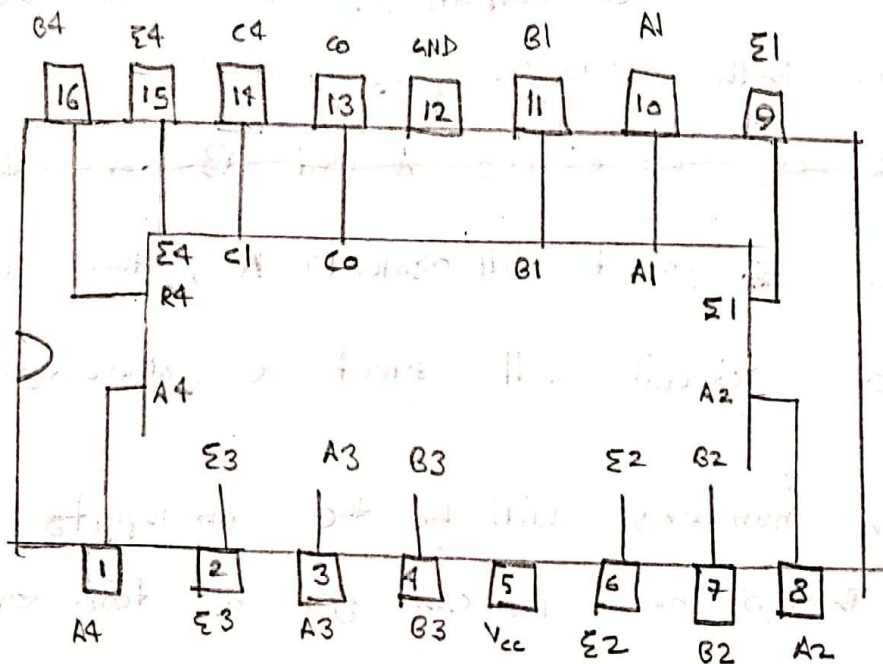
- I. MUX 74153. (4:1). — 2 x 2
- II. Adder 7483. (2bit). — 1 x 1
- III. Wires.

Ⓐ Experimental Setup :



MUX 74153

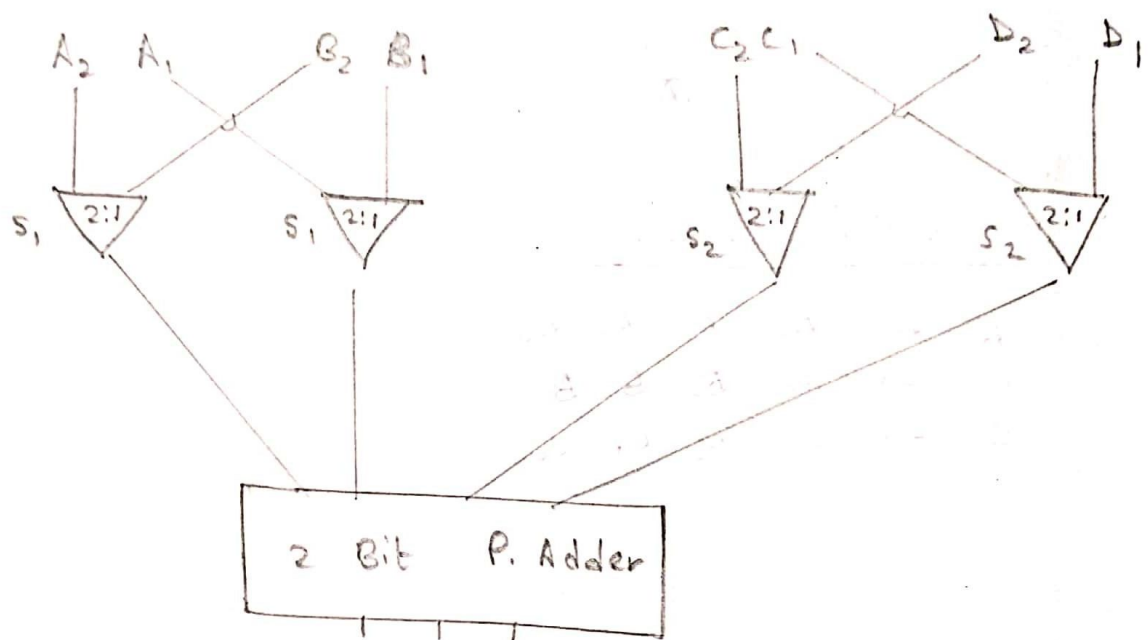
A, B = Select Pins



Adder 7483

(pro).

⑧ Experimental Setup:



Results:

Truth Table

S	S ₁	S ₂	Output	A/B		C/D		Sum	Carry
				Input 1 (MSB)	Input 1 (LSB)	Input 2 (MSB)	Input 2 (LSB)		
	0	0	A+C	0	0	1	1	11	0
	0	1	A+D	0	1	1	0	11	0
	1	0	B+C	1	0	0	0	10	0
	1	1	B+D	0	0	1	1	11	0

Discussions:

First of all, we had to use ~~4:1~~ ^{2:1} mux as 2:1 mux to get our desired outputs. To do so, we used one single pin for both Selector pins.

~~For S₁ = 0, we choose A and B otherwise.~~

If S₁ = 0, circuit will select A, otherwise B.

If S₂ = 0, circuit will select C, otherwise D.

Now, chosen numbers will be the inputs for 2 bit PA. That's how we can get a two or three bit binary number from a 4-bit number's system adding just 2-bit numbers.

$$\begin{aligned}
 40. \quad 2 - 2\cos\theta &= 4 \\
 2\cos\theta &= -2 \\
 \cos\theta &= -1 \\
 \theta &= \pi
 \end{aligned}$$

Ⓐ Question's answer:

$A \rightarrow A_3 \ A_2 \ A_1 \ A_0$
 $B \rightarrow B_3 \ B_2 \ B_1 \ B_0$
 $C \rightarrow C_3 \ C_2 \ C_1 \ C_0$

←
MSB

